

Mentoring Plan

The project supports one postdoctoral researcher, co-advised by PI Napolitano and Co-PI Wauthier (SAR processing and Bayesian fusion), and one PhD student, advised by Co-PI Busse and co-mentored by all PIs (mass balance modeling and community co-development). All PIs are at Penn State, enabling continuous cross-group engagement rather than periodic visits. Both mentees receive cross-disciplinary training spanning SAR imagery processing, hydrological mass balance, probabilistic structural assessment, and community-engaged research. The plan will be updated annually based on mentee feedback and assessed through progress reports and professional achievements.

Regular meetings and cross-disciplinary co-mentoring: PI Napolitano and Co-PI Wauthier will hold weekly meetings with the postdoc; Co-PI Busse weekly with the PhD student. Monthly cross-team meetings ensure integration across the SAR, hydrological, and co-design components. Both mentees attend lab meetings across all three groups: the postdoc joins Busse's mass balance meetings; the PhD student works with Napolitano on fragility surfaces and attends Wauthier's SAR sessions. Both participate in Napolitano's annual community co-development workshops in Montpelier.

Career counseling and professional development: Mentees will present at conferences (AGU, ASCE EMI, Association for Preservation Technology) and submit to peer-reviewed journals. Each PI provides career counseling on academic, industry, and government pathways.

Grant writing and scholarly communication: Training in preparing NSF proposals, publications, and presentations through Penn State's Graduate Writing Center. PI Napolitano will share materials from her graduate research methods course.

Teaching and mentoring skill development: Mentees will assist in developing project-related course content. The PhD student will be encouraged to pursue the Penn State CIRTLL Certificate and will receive mentorship on supervising undergraduates through the Penn State Undergraduate Research and Fellowships Mentoring program.

Undergraduate research: Two undergraduates per year participate through the BESURE program at Penn State, with recruitment prioritizing underrepresented groups via the WISER and MURE programs.

Diversity training and responsible practices: All mentees participate in DEI training at Penn State and receive responsible conduct of research training through Penn State's SARI program, covering research ethics, data management, and intellectual property.